

Math 243: Homework 05

Due Monday, September 28

Problem 1. Evaluate the following antiderivative

$$\int \sin^5(x) \cos^4(x) dx.$$

Problem 2. The distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) in $\mathbb{R}^3$ is

$$\text{distance} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

Using your own words, explain why this equation holds. Please include a sketch in your explanation.

Problem 3. Let $\mathbf{v} = -1\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$.

- (a) Compute $|\mathbf{v}|$.
- (b) Normalize $\mathbf{v}$.

Problem 4. Consider the parametric curve

$$\begin{cases} x(t) = \cos(t) & : \\ y(t) = \cos(3t) & : \end{cases}, \quad -\infty < t < \infty.$$

- (a) Sketch the curve.
- (b) For what values of t is the tangent to the curve horizontal? For what values is it vertical?
- (c) What are the (x, y) -coordinates of the points on the curve where the tangent to the curve is horizontal?

Problem 5. Evaluate the following integral:

$$\int_2^4 \frac{x+2}{x^2+3x-4} dx$$

Problem 6. The Cartesian equation of a sphere in $\mathbb{R}^3$ is

$$(x-h)^2 + (y-k)^2 + (z-\ell)^2 = r^2, \tag{1}$$

where (h, k, ℓ) is the center of the sphere, and $r > 0$ is the radius.

- (a) Show that

$$x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$$

is the equation of a sphere by rewriting it to be of the form shown in Eq. (1). (Complete the square)

- (b) What is the center and radius of the sphere?

Problem 7.

- (a) Use the identity $\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$ to show that

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}.$$

- (b) Using $\sin^2(x) + \cos^2(x) = 1$, use your answer to part (a) to conclude that

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}$$

Problem 8. Consider the parametric curve

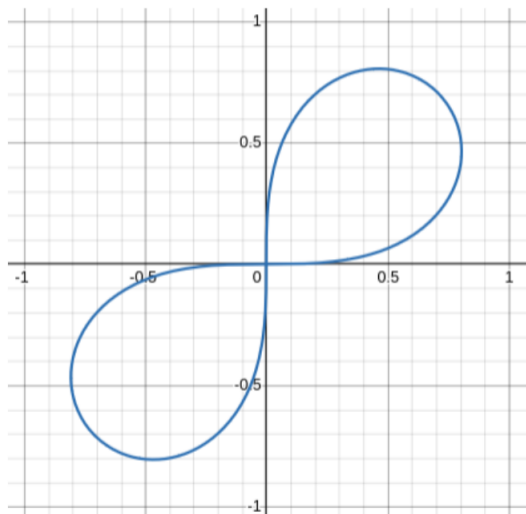
$$r = e^\theta, \quad 0 \leq \theta \leq 2\pi.$$

- (a) Sketch the curve.
- (b) Find the arc length of the portion of the curve in the upper half plane (i.e., where $y \geq 0$).

Problem 9. Consider the polar graph

$$r^2 = \sin(2\theta), \quad 0 \leq \theta \leq \frac{\pi}{2} \text{ and } \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}.$$

which has the following graph



Find the area of the enclosed region by computing the area of the top right petal.

Problem 10. Consider the polar function

$$r = \cos(\theta) + \sin(\theta), \quad 0 \leq \theta \leq \pi.$$

- (a) Using the formula (15) from lecture 10, list all points (r, θ) such that the tangent line is horizontal or vertical.
- (b) Convert the equation $r = \cos(\theta) + \sin(\theta)$ to Cartesian coordinates.
- (c) Using the area theorem from lecture 10, compute the area of the region bounded by the curve.
- (d) Sketch a graph of the curve and explain your answers to parts (a)-(c) geometrically.

Problem 11. Determine whether the given vectors are orthogonal, parallel, or neither:

- (a) $\langle 9, 3 \rangle$ and $\langle -2, 6 \rangle$
- (b) $\langle 4, 5, -2 \rangle$ and $\langle 3, -1, 5 \rangle$
- (c) $-8\mathbf{i} + 12\mathbf{j} + 5\mathbf{k}$ and $6\mathbf{i} - 9\mathbf{j} - 3\mathbf{k}$
- (d) $\langle 3, -1, 3 \rangle$ and $\langle 5, 9, -2 \rangle$